

Vincent Dominguito Abella *Software Developer*

✉ vinceabella05@gmail.com

☎ (+63) 9453869447

🌱 roastedbeans

📍 Cebu City, Philippines

🌐 <https://www.linkedin.com/in/vinch05/>

Profile

I am Vince, a software engineer and cybersecurity researcher specializing in independent web-based projects. I have experience leading and contributing to development teams, building technical expertise and collaborative skills. I focus on creating solutions that deliver meaningful impact.

Professional Experience

2024/03 – 2025/08 New Hudson, Michigan, United States	Software Engineer (Full-Stack Web Developer) <i>Mod Technologies LLC</i> - Developed frontend and backend features for the business website using Next.js, TypeScript, and Tailwind CSS with responsive design across platforms. - Migrated legacy React.js codebase to Next.js framework. - Integrated GraphQL queries into the frontend application.
2024/02 – 2024/05 Cebu City, Philippines	Digital Solutions Specialist (Internship) <i>CodeLikeUs Techonologies, Inc.</i> - Developed automation script for inventory tracking system. - Provided network setup solutions for in-store devices.
2023/09 – 2024/07 Cebu City, Philippines	Web Developer <i>Today's Carolinian</i> 🌐 Developed the frontend of the intramurals sports scoring system using Nuxt.js, PrimeVue, and Tailwind CSS with responsive design across platforms.
2024/11 – Present Germany	Web Developer <i>Techflow.ai</i> 🌐 - Developed full-stack no-code dashboard using weweb.io 🌐 to manage internal payment system. - Built full-stack e-commerce website using weweb.io 🌐, Make automations, and OpenAI.
2023/10 – 2023/12 Cebu City, Philippines	Freelance Web Developer <i>Virgorus</i> - Built admin interface using Next.js, TypeScript, NextUI, and Tailwind CSS with responsive design for multiple platforms. - Resolved application bugs and maintained codebase stability.
2023/07 – 2023/09 Cebu City, Philippines	Software Developer <i>Purple Cow</i> 🌐 - Developed frontend architecture using React.js, TypeScript, Tailwind CSS, and Material UI with responsive design across platforms. - Designed UI/UX in Figma.
2024/08 – Present Seoul, South Korea	Cybersecurity Researcher <i>Mobile Internet Security Lab</i> 🌐 - Developed testbed to generate datasets for integrated authentication in South Korea's financial service and test intrusion detection systems. - Built 5G global roaming testbed in Kubernetes cloud environment using Open5GS and PacketRusher. - Developed Android application to collect signal strength and layer 3 protocol data from 4G and 5G networks on rooted Android 11+ devices.

- Built Android application for detecting false base stations using specification-based and AI-based intrusion detection systems.

2025 – Present
Seoul, South Korea

Web Chair
<https://ebision.ifip-wg84.org/>

- Developed and designed website for EBISION IFIP WG 8.4 international conference.
- Developed the conference management system with analytics dashboard, paper submissions, payments, attendance, online poster, email notifications, certification generation.

Education

present Cebu City, Philippines	Bachelor of Science in Computer Engineering <i>University of San Carlos</i>
2024/08 – Present Seoul, South Korea	Masters of Science in Cybersecurity <i>Kookmin University</i>

Skills

Languages TypeScript, JavaScript, Python, C, C#, PHP	Frontend React.js, Next.js, Nuxt.js, Tailwind CSS, Material UI, PrimeVue
Backend Node.js, Express.js, Laravel, GraphQL (Apollo Client/Server, Resolvers, Schema Design)	Mobile Native Android, Rooted Devices, Qualcomm Baseband, Security Policies, Chaquopy
Networking and Security • Networks: Ethernet, IP Connectivity, Subnetting, IPv4/IPv6, Network Protocols, Switching • Telecom: 4G/5G Networks, LTE, Kubernetes, Telecom Core Networks • Security: Intrusion Detection Systems, AI-based Threat Detection, False Base Station Detection	Database MySQL, MongoDB, Postgres, Supabase
Core Concepts Data Structures, Algorithms, Embedded Systems, Computer Organization and Architecture	Tools and Methodologies Git, GitHub, Trello, Notion, Agile, Unity, Simulation Environments, AI Model Performance Testing

Certificates

The Fundamentals of Digital Marketing Issued by Google Digital Garage Issued on April 2023 Certificate ID: HB5 AE9 DBN	CCNA: Introduction to Networks Issued by Cisco Issued on January 2023
Graph Developer - Associate Apollo Fundamental understanding of GraphQL, schema, useQuery hooks, and Apollo tools	

Projects

2023/12 – present Cebu City, Philippines	Project Taytayan (<i>Next.js, Flowbite-react, Firebase Firestore</i>) Google Solutions Challenge 2024 • Built web application centralizing training, workshops, and employment opportunities for out-of-school youth.
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	• Developed admin dashboard with CRUD operations using Firebase Firestore. • Debugged and resolved issues throughout development.
2023/10 – present Cebu City, Philippines	EZ-Forms <i>(Next.js, OpenAI, ShadCN UI)</i> • Developed frontend for full-stack application that automates the generation of multiple activity forms, reducing manual form creation time. • Implemented dynamic form generation logic allowing users to create customized activity forms based on configurable templates.
2023/09 – present Surigao City, Philippines	BOOKIP <i>(React.js, Node.js, Express, MySQL, ShadCN UI)</i> • Developed full-stack booking system for small business lodging houses with automated reservation management. • Built financial tracking module with transaction history, cashflow monitoring, and tax calculation. • Implemented responsive design across platforms.
2023/04 – 2023/12 Cebu City, Philippines	Development of the Simulation for Grocery Shopping Optimization with Machine Learning-Driven Sorting Algorithm <i>(Unity Game Engine, C#)</i> • Developed top-down supermarket simulation integrated with ML-driven sorting algorithm via socket communication. • Utilized Python libraries to generate performance graphs validating algorithm efficiency across multiple shopping scenarios.
2024 Cebu City, Philippines	Chelo  <i>Event Organization Platform</i> • Developed full-stack event organization platform enabling organizers to create custom registration forms and process participant payments. • Built using Next.js, Tailwind CSS, shadcn, and Supabase.
2025/12 – Present Seoul, South Korea	Archway <i>Conference Management Platform</i> • Developed conference management system providing support for research organizations in handling the conference workflows. • The system integrates paper submissions, review system, payments, analytics, email notifications, QR-based attendance, certificate generation, session scheduling.
2026 – Present Cebu City, Philippines	Xylvir Technologies <i>Xylvir Technologies Portfolio</i> • Developed the website for xylvir technologies providing the portfolio for all the projects

Organizations

2023/09 – present Cebu City, Philippines	Today's Carolinian <i>Web Developer</i>
2022 – 2023 Cebu City, Philippines	USC - Computer Engineering Council <i>Councilor</i>
2023/08 – present Cebu City, Philippines	CoDE - University of San Carlos <i>Events Coordinator, Frontend Web, VEX</i>
Present Cebu City, Philippines	OWASP Cebu  <i>Co-lead</i> OWASP Cebu City Chapter is a local community dedicated to web application security.

Awards

2024	1st Place - GDSC Visayas Solve 2024 <i>GDSC Visayas</i>
2023	1st Place - Regional ICpEP Quiz Bowl Competition <i>Institute of Computer Engineers of the Philippines</i>
2024/07	1st Place - Startup Weekend Bohol 2024 <i>Startup Weekend Bohol</i> A 54-hour event for sharing ideas, forming teams, building products, and launching businesses.
2024/12	Best Poster Paper <i>Mobile Internet Security</i> The 8th International Conference on Mobile Internet Security

Publications

2026/02	Benchmarking Deep Learning Architectures for Real-Time Intrusion Detection in Kubernetes-Orchestrated 5G Core Networks ↗ <i>Research Briefs on Information and Communication Technology Evolution</i> , 12, 57–66. https://doi.org/10.64799/rebict.e.V12.3 ↗
2025/11	Enhancing Roaming Security in Cloud-Native 5G Core Network through Deep Learning-Based Intrusion Detection System ↗ CMES - Computer Modeling in Engineering and Sciences, Volume 145, Issue 2, 2025, Pages 2733-2760, ISSN 1526-1492, https://doi.org/10.32604/cm.e.2025.072611 ↗
2024/08	Machine Learning-Assisted Dynamic Proximity-Driven Sorting Algorithm for Supermarket Navigation Optimization: A Simulation-Based Validation ↗ <i>Future Internet</i> 2024, 16, 277. https://doi.org/10.3390/fi16080277 ↗
2026/02	MODI: On-Device Preprocessing of Qualcomm Diagnostic Logs for Cellular Protocol Analysis on Android ↗ <i>Journal of Internet Services and Information Security (JISIS)</i> , volume: 16, number: 1 (February-2026), pp. 612-623. DOI: 10.58346/JISIS.2026.I1.034
2025/12	Extending Open5GS for IIoT: Performance Evaluation of the EAP-AKA'Integration for 5G Standalone Non-Public Networks ↗ <i>Research Briefs on Information and Communication Technology Evolution</i> . 11, (Dec. 2025), 199–208. DOI: https://doi.org/10.64799/rebict.e.V11.13 ↗ .
2026/06	SMDEVC: Specification-Based Misbehavior Detection for Electric Vehicle Charging Stations ↗ <i>Applied Science</i> 2026, 16(11), 5605; https://doi.org/10.3390/app16115605 ↗

References

Ma. Luisa Guyano, *School Principal*, Caraga Regional Science High School
luisaguyano.lg@gmail.com, 0939 905 8510

Jhury Kevin Lastre, *Mid-level Software Engineer*, Lanex Philippines
jhurylastre432@gmail.com, 0949 186 3088

Rovic Ira Correos, *Assistant Manager (Bank Officer I)*, Development Bank of the Philippines DBP
rovicira.correos@gmail.com, 0963 927 1221

Ilsun You, *Professor*, Kookmin University
ilsunu@gmail.com

Philip Astillo, *Professor*, University of San Carlos
pvbastillo@gmail.com